NewsBot Intelligence System

Group Reflective Journal

ITAI 2373 — Natural Language Processing | Spring 2026

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1. Team Collaboration Analysis

This project was completed as a solo effort, which honestly made it both harder and more rewarding at the same time. Without teammates to divide up the modules, I had to work through every single component myself — from data loading all the way to the final integration class. There were moments where I genuinely wished I had someone to bounce ideas off of, especially when debugging the feature matrix in Module 7 or figuring out why spaCy kept throwing errors in Module 8.

That said, doing it solo forced me to actually understand every part of the pipeline rather than just owning one module. I couldn't hand off a broken function to someone else and move on — if something was wrong, I had to figure it out. Looking back, that pressure made me a lot more confident in the material. I managed my time by tackling one module per session and keeping detailed comments in my notebook so I could pick up where I left off without losing context.

2. Technical Integration Challenges

The two modules that gave me the most trouble were Module 7 (Classification) and Module 8 (Named Entity Recognition). For Module 7, the main issue was building the combined feature matrix. When I tried to stack TF-IDF features with POS proportions and emotion scores using scipy's hstack, I kept getting a ValueError about unsupported dtypes. It turned out that some of my columns — like pos_results and dominant_emotion — were storing Python dictionaries and strings, not floats. The fix was writing a safe_sparse() helper function that forces everything to numeric and drops non-numeric columns before building the sparse matrix. Simple in hindsight, but it took a while to track down.

Module 8 had a different kind of challenge. The spaCy NER model works well overall, but on news text it sometimes misses entities or misclassifies them — tagging a country as an organization, for example. I also ran into performance issues trying to run NER on all 1,490 articles in one go on Colab's free tier. I solved that by truncating each article to 2,000 characters before passing it to the model, which cut processing time significantly without losing much entity coverage since most key entities appear early in news articles anyway. Integrating all eight modules into the final NewsBotIntelligenceSystem class was its own challenge — making sure every method used consistent column names and that the classifier received the same feature shape it was trained on took several rounds of debugging.

3. Business Value Assessment

Before starting this project I understood NLP conceptually but didn't have a clear picture of how it translates into something actually useful. Building NewsBot changed that. The classification system alone — hitting around 97% accuracy on five categories — is genuinely production-ready for a media monitoring use case. A company that needs to sort thousands of incoming news articles per day could deploy something like this and replace a significant amount of manual editorial work.

The sentiment and NER combination also opened my eyes to how financial firms and PR agencies use NLP. Being able to automatically flag that politics coverage skews negative, or that a specific company is mentioned alongside certain sentiment patterns, is exactly the kind of signal that risk teams and brand managers pay for. The TF-IDF signature term analysis was another insight I hadn't expected — seeing that "chancellor", "election", and "parliament" dominate politics while "software", "broadband", and "mobile" dominate tech made the vocabulary differences between categories feel really concrete.

4. Individual Contributions

Module	Component	Contribution
Module 1	Business Context	Defined use case, target users, value proposition
Module 2	Preprocessing	Built clean_text() and preprocess() pipeline
Module 3	TF-IDF	Vectorizer tuning, category term analysis, heatmap
Module 4	POS Tagging	POS grouping, noun/verb ratio, bigram patterns
Module 5	Syntax Parsing	Dependency parsing, SVO extraction, tree depth
Module 6	Sentiment	VADER scoring, 8-emotion lexicon, visualizations
Module 7	Classification	5 models, safe_sparse() fix, error analysis
Module 8	NER	Entity extraction, co-occurrence, cross-category analysis
Integration	Full System	NewsBotIntelligenceSystem class, batch pipeline

5. Future Enhancements

The most impactful improvement I would make with more time is replacing the current classification models with a fine-tuned BERT model. The Linear SVM hitting 97% accuracy is impressive for a traditional ML approach, but transformer-based models like BERT would handle ambiguous articles much better — the ones that sit between politics and business, or between tech and entertainment. The contextual embeddings would also benefit the NER component significantly, reducing the misclassification errors I noticed with the spaCy small model.

Beyond the model upgrade, I'd want to connect the system to a live news API like NewsAPI or RSS feeds so it processes articles in real time rather than a static dataset. Adding a Streamlit dashboard on top of that would make it genuinely usable by non-technical users — someone in a marketing or PR role could paste in a URL and instantly get a full intelligence report without touching any code. I'd also expand the entity relationship mapping to build a proper knowledge graph showing how people, organizations, and locations are connected across articles over time.

6. Professional Development Impact

This project genuinely changed how I think about NLP as a career skill. Going in, I thought of text classification as something you just call a function for. Coming out, I understand the full pipeline — why preprocessing decisions affect downstream accuracy, why feature engineering matters even when you have a strong TF-IDF baseline, and how combining multiple signal types (sentiment, POS, entities) improves robustness beyond what any single feature set can achieve on its own.

The part that surprised me most was how high the classifier accuracy ended up being — around 97% on a five-category problem using a relatively simple Linear SVM. I went in expecting to need something much more complex to get decent results, and realizing that a well-built traditional ML pipeline can outperform expectations was a useful lesson in not over-engineering solutions. This notebook is something I'm genuinely proud to put in my portfolio. It demonstrates the full NLP workflow in a way that's readable, well-documented, and actually functional — which is exactly what I'd want a potential employer to see.